

Results Of CARP over SST-2 From Hugging Face Library

SST2 data-set

- A well-known benchmark for sentiment analysis that classifies sentences into positive or negative sentiment categories
- The corpus is based on the dataset introduced by Pang and Lee (2005) and consists of 11,855 single sentences extracted from movie reviews

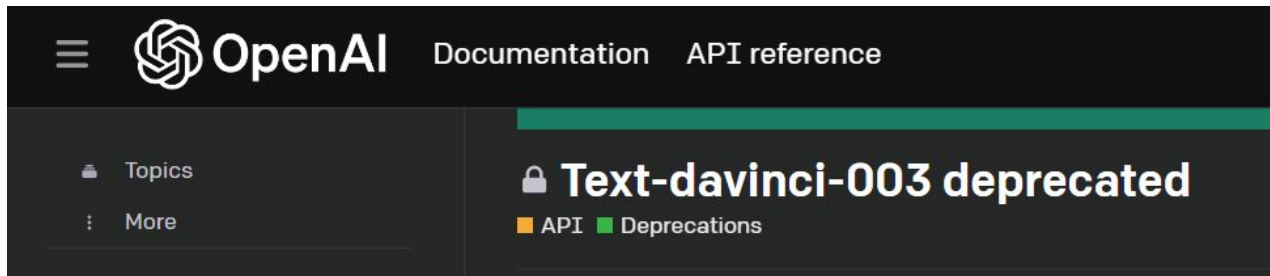
0	hide new secretions from the parental units	0 negative
1	contains no wit , only labored gags	0 negative
2	that loves its characters and communicates something rather beautiful about human nature	1 positive
3	remains utterly satisfied to remain the same throughout	0 negative
4	on the worst revenge-of-the-nerds clichés the filmmakers could dredge up	0 negative
5	that 's far too tragic to merit such superficial treatment	0 negative
6	demonstrates that the director of such hollywood blockbusters as patriot games can still turn out a small , personal film with an emotional wallow...	1 positive
7	of saucy	1 positive

Initial Approach from the Paper

- Model Used: The original CARP method used OpenAI's **text-davinci-003 model**.
- Why Change Was Needed: text-davinci-003 is now deprecated, so we explored alternatives.

Alternative

- Our Chosen Model: **GPT-3.5-turbo-instruct**
- A close alternative to text-davinci-003, designed for instructive tasks.
- Purchased openAi credits worth \$23



CARP prompting Technique as per the Paper

This is an overall sentiment classifier for movie reviews.

First, list CLUES (i.e., keywords, phrases, contextual information, semantic relations, semantic meaning, tones, references) that support the sentiment determination of input..

Second, deduce the diagnostic REASONING process from premises (i.e., clues, input) that supports the INPUT sentiment determination (Limit the number of words to 130).

Third, based on clues, reasoning and input, determine the overall SENTIMENT of INPUT as Positive or Negative.

INPUT: press the delete key

CLUES: delete key

REASONING: The phrase "delete key" implies an action of removing something, which could be interpreted as a negative sentiment.

SENTIMENT: Negative ✓

(c) CARP Zero-shot

Implementation Details

```
def classify_text(input_text):  
    prompt = f"""  
        This is an overall sentiment classifier for movie reviews.  
        First, present CLUES (i.e., keywords, phrases, contextual information, semantic relations, semantic meaning,  
        tones, references) that support the sentiment determination of input.  
        Second, deduce a diagnostic REASONING process from premises (i.e., clues, input) that supports the sentiment  
        determination (Limit the number of words to 130).  
        Third, determine the overall SENTIMENT of INPUT as Positive or Negative considering CLUES, the REASONING  
        process, and the INPUT.  
        INPUT: {input_text}  
        """>  
  
    response = openai.Completion.create(  
        engine="gpt-3.5-turbo-instruct",  
        prompt=prompt,  
        max_tokens=150,  
        temperature=0.7  
    )
```

Example Review

```
37 # Example usage
38 text_input = "but like most rabbits , it seems to lack substance "
39 full_response, sentiment_value = classify_text(text_input)
40 print("Full Response:", full_response)
41 print("Sentiment Value:", sentiment_value)
42
```

Full Response:

CLUES:

1. "like most rabbits" - suggests a comparison to other rabbits
2. "lack substance" - implies a lack of depth or quality

REASONING PROCESS:

This sentence compares something to "most rabbits", which can be interpreted as being common or average. The fo

OVERALL SENTIMENT:

Negative

Sentiment Value: 0

Conclusions when tested on sst2-test data

- Contains 1820 movie reviews
- Time taken around 42 minutes
- Accuracy = 72.76 % (Low when compared to the results from the paper)

```
1 accuracy = np.sum([1 if l1==l2 else 0 for l1,l2 in zip(df['new_label']  
2 accuracy/len(df)
```

```
0.727622185612301
```

Modified Approach for Prompt

Our Modified Prompt Technique

Instead of a single prompt for the entire sentiment analysis of **each review** , we divided it into sequential stages, Breaking down single prompt request for each review into **THREE** separate prompt requests for final classification.

1. **Clues Prompt:** First, asked the model to identify clues about sentiment.
2. **Reasoning Prompt:** Next, requested reasoning based on identified clues.
3. **Sentiment Prompt:** Finally, asked for the sentiment determination

Implementation Details

```
# Step 1: Identify Clues
clues_prompt = (
    f"This is an overall sentiment classifier.\n"
    f"First, list CLUES (i.e., keywords, phrases, contextual information, semantic relations, "
    f"semantic meaning, tones, references) that support the sentiment determination of the input.\n"
    f"Input: {input_text}\n"
    f"Clues:"
)

# Step 2: Diagnostic Reasoning
reasoning_prompt = (
    f"Based on the input and clues, deduce the diagnostic reasoning process from premises "
    f"(i.e., clues and input) that supports the sentiment determination of the input.\n"
    f"Input: {input_text}\n"
    f"Clues: {clues}\n"
    f"Reasoning:"
)

# Step 3: Final Sentiment Determination
sentiment_prompt = (
    f"Considering the clues and reasoning, determine the overall SENTIMENT of the input as Positive, "
    f"or Negative.\n"
    f"Input: {input_text}\n"
    f"Clues: {clues}\n"
    f"Reasoning: {reasoning}\n"
    f"Sentiment:"
)
```

Example Review

```
80 # Example usage
81 input_text = "its mysteries are transparently obvious , and it 's too slowly paced to be a thriller"
82 result = classify_text_improved(input_text)
83
84 print("Final Sentiment:", result)
85
```

Clues Response: mysteries, transparently obvious, slowly paced, thriller

Reasoning Response:

The input mentions that the subject matter at hand is "mysteries," and that these mysteries are "transparently obvious." This su

Sentiment Determination Response: Negative

Final Sentiment: 0

Conclusions when tested on sst2-test data

Accuracy: 89.34% , much better than previous approach.

Time taken : more than 1 hour

```
5 accuracy = np.sum([1 if l1==l2 else 0 for l1,l2 in zip
6
7 accuracy/len(df)
```



0.8934651290499726

Possible Reasons of Low Accuracy for First Approach

Much better performance of the davinci-003 model used in paper reported by users in the openAI Forum



Rafs

Feb 25

The emotional intelligence of DaVinci-003 is way higher than gpt-3.5-turbo. Is there any way of still using it?

See paper: Emotional intelligence of Large Language Models Xuena Wang, Xueting Li, Zi Yin, Yue Wu, Jia Liu



styx

Jul 2023

I hope there is a way to keep at least the text-davinci-003 model available in the future!

It works **so much better than GPT-3 or 4** to follow specific linguistic instructions to create articles!

The GPT models will just ignore many of the instructions. I've compared with many prompts and thousands of articles.

The same prompts generated with davinci-003 even pass AI content detectors while the same prompt used with any GPT model won't, but that is not the point to keep it around, just another point.

The quality of generated text from davinci is a lot better and higher quality than any of the chatGPT models produce.

So please OpenAI, save davinci-003 and keep it in the API 😊

Possible Strategies to Increase Performance

- Use a better model
- Accuracy averaged
Over 5 runs as
Evaluated in paper
(Not sure about this)

Models overview

The OpenAI API is powered by a diverse set of models with different capabilities and price points. You can also make customizations to our models for your specific use case with [fine-tuning](#).

MODEL	DESCRIPTION
GPT-4o	Our high-intelligence flagship model for complex, multi-step tasks
GPT-4o mini	Our affordable and intelligent small model for fast, lightweight tasks
o1-preview and o1-mini	Language models trained with reinforcement learning to perform complex reasoning.
GPT-4 Turbo and GPT-4	The previous set of high-intelligence models
GPT-3.5 Turbo	A fast, inexpensive model for simple tasks
DALL·E	A model that can generate and edit images given a natural language prompt
TTS	A set of models that can convert text into natural sounding spoken audio
Whisper	A model that can convert audio into text
Embeddings	A set of models that can convert text into a numerical form
Moderation	A fine-tuned model that can detect whether text may be sensitive or unsafe
Deprecated	A full list of models that have been deprecated along with the suggested replacement

Results as shown in Paper

	SST-2	AGNews	R8	R52	MR	Average
<u>Supervised Methods</u>						
RoBERTa-Large (Liu et al., 2019)	95.99	95.55	97.76	96.42	91.16	95.38
DeBERTa (He et al., 2020)	94.75	95.32	98.33	96.32	90.19	94.99
RoBERTa-GCN (Lin et al., 2021)	95.80	95.68*	98.2	96.1	89.7	95.10
XLNet (Yang et al., 2019)	96.10*	95.55	-	-	-	-
VLAWE (Ionescu and Butnaru, 2019)	-	-	-	-	93.3*	-
GCN-SB (Zeng et al., 2022)	-	-	98.53*	96.35*	87.59	-
<u>Zero-shot Setting</u>						
Vanilla (Brown et al., 2020)	91.55	90.72	90.19	89.06	88.69	90.04
CoT (Kojima et al., 2022)	92.11	91.25	90.48	91.24	89.37	90.89
CARP	93.01	92.60	91.75	91.80	89.94	91.82

Table 2: Accuracy performances of different settings on benchmarks. We report mean and standard deviation results over 5 runs. The GPT-3 denotes `text-davinci-003`. In few-shot experiments, we sample 16 annotated examples ($k=16$) for every test instance. * indicates previous state-of-the-art results. "MJ Vote" is short for majority vote. "WP Vote" denotes weighted probability vote.

Evaluation metrics for our podcast case.

END